

Unit 1: Introduction to Artificial Intelligence & Intelligent Agents

What is Artificial Intelligence?

Definition

- **AI:** Making machines think and act like humans
- **Goal:** Create smart machines that can solve problems
- **Main Idea:** Computers that can learn, reason, and make decisions

Types of AI

1. Narrow AI (Weak AI)

- Designed for specific tasks only
- Examples: Siri, Google Translate, Chess programs
- Current technology level

2. General AI (Strong AI)

- Human-level intelligence in all areas
- Can think and reason like humans
- Still in research phase

3. Super AI

- More intelligent than humans
- Theoretical concept
- Future possibility

History of AI

Important Milestones

- **1950:** Alan Turing creates Turing Test
- **1956:** Dartmouth Conference - AI field officially born
- **1960s:** First AI programs and expert systems
- **1970s:** Knowledge-based systems development
- **1980s:** Machine learning concepts emerge
- **1990s:** Internet and data explosion
- **2000s:** Big data and improved algorithms

- **2010s:** Deep learning revolution
- **Present:** AI everywhere - phones, cars, homes

Applications of AI

1. Healthcare

- **Medical Diagnosis:** X-ray analysis, disease detection
- **Drug Discovery:** Finding new medicines faster
- **Surgery Robots:** Precise operations
- **Patient Monitoring:** Continuous health tracking

2. Transportation

- **Self-Driving Cars:** Tesla, Google cars
- **Traffic Management:** Smart traffic lights
- **Route Optimization:** GPS navigation
- **Predictive Maintenance:** Vehicle servicing

3. Finance

- **Fraud Detection:** Unusual transaction identification
- **Credit Scoring:** Loan approval decisions
- **Algorithmic Trading:** Automated stock trading
- **Customer Service:** Chatbots for banking

4. Entertainment

- **Game Playing:** Chess, Go, video games
- **Content Recommendation:** Netflix, YouTube suggestions
- **Music Creation:** AI-generated songs
- **Virtual Reality:** Immersive experiences

5. Education

- **Personalized Learning:** Adaptive study programs
- **Automated Grading:** Essay and answer evaluation
- **Language Learning:** Translation and pronunciation
- **Virtual Tutors:** 24/7 learning assistance

Intelligent Agents

What is an Intelligent Agent?

- **Definition:** A system that perceives environment and takes actions
- **Goal:** Maximize performance or achieve objectives
- **Key Feature:** Acts autonomously (independently)

Agent Components

1. Sensors:

- Input devices that observe environment
- Examples: Camera, microphone, temperature sensor

2. Actuators:

- Output devices that perform actions
- Examples: Motors, speakers, display screen

3. Agent Program:

- Software that processes input and decides actions
- Brain of the agent

Agent Function

- **Input:** Sequence of perceptions
- **Output:** Action to be performed
- **Mapping:** Perception → Action

Types of Intelligent Agents

1. Simple Reflex Agents

- **Behavior:** Act based on current perception only
- **Rule:** If condition then action
- **Example:** Thermostat (if temp > 25°C then turn on AC)
- **Limitation:** No memory of past events

2. Model-Based Reflex Agents

- **Feature:** Maintain internal state/memory
- **Advantage:** Remember previous situations

- **Example:** Robot vacuum that maps the room
- **Components:** Current state + sensors + actuators

3. Goal-Based Agents

- **Feature:** Have specific goals to achieve
- **Decision Making:** Choose actions that lead to goals
- **Example:** GPS navigation (goal: reach destination)
- **Planning:** Think about future consequences

4. Utility-Based Agents

- **Feature:** Maximize expected utility (happiness/satisfaction)
- **Decision:** Choose action with highest utility value
- **Example:** Stock trading bot (maximize profit)
- **Advantage:** Handle conflicting goals

5. Learning Agents

- **Feature:** Improve performance through experience
- **Components:**
 - Learning Element: Modifies performance
 - Performance Element: Selects actions
 - Critic: Provides feedback
 - Problem Generator: Suggests new experiences
- **Example:** Recommendation systems that learn user preferences

Agent Environment Types

1. Observable vs Partially Observable

- **Fully Observable:** Agent sees complete environment state
 - Example: Chess game (see entire board)
- **Partially Observable:** Limited visibility
 - Example: Self-driving car (can't see around corners)

2. Deterministic vs Stochastic

- **Deterministic:** Same action always produces same result

- Example: Calculator ($2+2$ always equals 4)
- **Stochastic:** Random/unpredictable outcomes
 - Example: Weather prediction, stock market

3. Episodic vs Sequential

- **Episodic:** Current action doesn't affect future episodes
 - Example: Email classification
- **Sequential:** Current action affects future choices
 - Example: Chess moves affect entire game

4. Static vs Dynamic

- **Static:** Environment doesn't change while agent thinks
 - Example: Crossword puzzle
- **Dynamic:** Environment changes during decision making
 - Example: Real-time strategy games

5. Discrete vs Continuous

- **Discrete:** Limited number of possible actions/states
 - Example: Chess (finite board positions)
- **Continuous:** Unlimited possibilities
 - Example: Driving (infinite steering angles)

6. Single-Agent vs Multi-Agent

- **Single-Agent:** Only one agent in environment
 - Example: Puzzle solving
- **Multi-Agent:** Multiple agents interact
 - Example: Multiplayer games, autonomous vehicles

Agent Architecture

Simple Agent Architecture

Environment → Sensors → Agent Program → Actuators → Environment

Detailed Architecture

1. **Perception:** Receive input from sensors
2. **Processing:** Analyze and decide using agent program
3. **Action:** Execute decision through actuators
4. **Feedback:** Observe results and learn

PEAS Description

Components

- **Performance Measure:** How to evaluate agent success
- **Environment:** The world agent operates in
- **Actuators:** Agent's action mechanisms
- **Sensors:** Agent's perception mechanisms

Example: Self-Driving Car

- **Performance:** Safety, speed, legality, comfort, profit
- **Environment:** Roads, traffic, weather, passengers
- **Actuators:** Steering, brake, accelerator, horn, display
- **Sensors:** Camera, radar, GPS, speedometer, microphone

Key Points for Exam

Important Definitions

- **AI:** Simulation of human intelligence in machines
- **Agent:** Entity that perceives and acts in environment
- **Rationality:** Acting to achieve best expected outcome
- **Autonomy:** Acting independently without human intervention

Classification Points

- **Agent Types:** Reflex → Model-based → Goal-based → Utility → Learning
- **Environment Types:** Observable, Deterministic, Episodic, Static, Discrete, Single-agent

Remember for Exams

1. **Turing Test:** Machine's ability to exhibit human-like intelligence

2. **Chinese Room:** Argument against machine consciousness
3. **Agent = Architecture + Program**
4. **Rational Agent:** Does right thing based on available information
5. **Performance Measure:** Objective criterion for agent success